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#include <iostream>
#include <string.h>

using namespace std;

struct Book {
    int id;
    char title[100];
    char author[100];
    char member[100];
    int price;
    int isIssued;
};

int main() {
    struct Book library[50];
    int count = 0;
    int choice = 0;

    while (choice != 5) {
        cout << "\n--- LIBRARY SYSTEM ---\n";
        cout << "1. Add Book\n";
        cout << "2. Issue / Return Book\n";
        cout << "3. Search Book\n";
        cout << "4. Display All Books\n";
        cout << "5. Exit\n";
        cout << "Enter choice: ";
        cin >> choice;

        if (choice == 1) {
            cout << "Enter Book ID: ";
            cin >> library[count].id;

            cout << "Enter Title: ";
            cin.ignore();
            cin.getline(library[count].title, 100);

            cout << "Enter Author: ";
            cin.getline(library[count].author, 100);

            cout << "Enter Price: ";
            cin >> library[count].price;

            strcpy(library[count].member, "None");

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library[count].isIssued = 0;

count++;
cout << "Book added successfully!\n";
}
else if (choice == 2) {
    int searchId;
    int found = 0;
    cout << "Enter Book ID to Issue/Return: ";
    cin >> searchId;

    for (int i = 0; i < count; i++) {
        if (library[i].id == searchId) {
            found = 1;
            if (library[i].isIssued == 0) {
                cout << "Enter Member Name: ";
                cin.ignore();
                cin.getline(library[i].member, 100);
                library[i].isIssued = 1;
                cout << "Book issued successfully!\n";
            } else {
                strcpy(library[i].member, "None");
                library[i].isIssued = 0;
                cout << "Book returned successfully!\n";
            }
            break;
        }
    }
    if (found == 0) {
        cout << "Book ID not found.\n";
    }
}
else if (choice == 3) {
    char query[100];
    int found = 0;
    cout << "Enter Title or Author to search: ";
    cin.ignore();
    cin.getline(query, 100);

    for (int i = 0; i < count; i++) {
        if (strstr(library[i].title, query) != NULL || strstr(library[i].author, query) != NULL) {
            cout << "\nFound Book:\n";
            cout << "ID: " << library[i].id << "\n";
            cout << "Title: " << library[i].title << "\n";
        }
    }
}

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        cout << "Author: " << library[i].author << "\n";
        if (library[i].isIssued == 1) {
            cout << "Status: Issued to " << library[i].member << "\n";
        } else {
            cout << "Status: Available\n";
        }
        found = 1;
    }
}
if (found == 0) {
    cout << "No matching book found.\n";
}
}
else if (choice == 4) {
    if (count == 0) {
        cout << "No records available.\n";
    } else {
        for (int i = 0; i < count; i++) {
            cout << "\nBook " << (i + 1) << ":\n";
            cout << "ID: " << library[i].id << "\n";
            cout << "Title: " << library[i].title << "\n";
            cout << "Author: " << library[i].author << "\n";
            cout << "Price: " << library[i].price << "\n";
            if (library[i].isIssued == 1) {
                cout << "Status: Issued to " << library[i].member << "\n";
            } else {
                cout << "Status: Available\n";
            }
        }
    }
}
else if (choice == 5) {
    cout << "Exiting system...\n";
}
else {
    cout << "Invalid choice, try again.\n";
}
}

return 0;
}

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